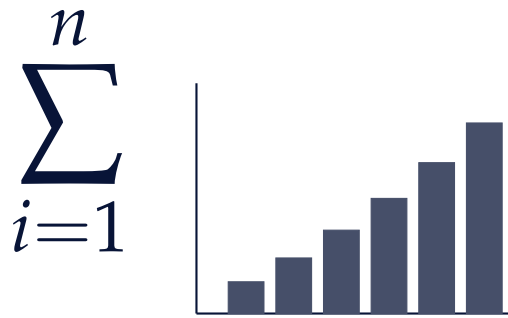




# Sum of Sequences Proof by Induction

## Practice Question Set

Proving Summation Formulae using the Principle of  
Mathematical Induction

4 Questions • Fully Worked Answers

**Instructions:** Attempt each proof yourself first. The full worked answer is given directly beneath each question, boxed in light green, so you can check your working immediately.

## Question 1

(a) Prove that for all  $n \in \mathbb{Z}^+$ :

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}.$$

(b) Hence find  $1^2 + 2^2 + 3^2 + \cdots + 100^2$ .

## Answer

(a) Let  $P_n$  be the proposition:  $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$  for all  $n \in \mathbb{Z}^+$ .

**Proof:** (By the principle of mathematical induction)

(1) If  $n = 1$ : LHS =  $1^2 = 1$  and RHS =  $\frac{1 \times 2 \times 3}{6} = 1$ .  $\therefore P_1$  is true.

(2) If  $P_k$  is true for  $k \in \mathbb{Z}^+$ , then

$$\sum_{i=1}^k i^2 = 1^2 + 2^2 + 3^2 + \cdots + k^2 = \frac{k(k+1)(2k+1)}{6}.$$

Thus

$$\begin{aligned} 1^2 + 2^2 + \cdots + k^2 + (k+1)^2 &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \\ &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \times \frac{6}{6} \\ &= \frac{(k+1)[k(2k+1) + 6(k+1)]}{6} \\ &= \frac{(k+1)(2k^2 + k + 6k + 6)}{6} \\ &= \frac{(k+1)(2k^2 + 7k + 6)}{6} \\ &= \frac{(k+1)(k+2)(2k+3)}{6} \\ &= \frac{[k+1]([k+1]+1)(2[k+1]+1)}{6}. \end{aligned}$$

$\therefore P_{k+1}$  is also true. Since  $P_1$  is true, and  $P_{k+1}$  is true whenever  $P_k$  is true,  $P_n$  is true for all  $n \in \mathbb{Z}^+$ . {principle of mathematical induction}

(b) Using  $P_{100}$ :  $1^2 + 2^2 + 3^2 + \cdots + 100^2 = \frac{100 \times 101 \times 201}{6} = 338\,350$ .

## Question 2

Use the principle of mathematical induction to prove that for all positive integers  $n$ :

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}.$$

## Answer

Let  $P_n$  be the proposition:  $\sum_{i=1}^n i = \frac{n(n+1)}{2}$  for all positive integers  $n$ .

**Proof:** (By the principle of mathematical induction)

(1) If  $n = 1$ : LHS = 1 and RHS =  $\frac{1(2)}{2} = 1$ .  $\therefore P_1$  is true.

(2) If  $P_k$  is true, then  $\sum_{i=1}^k i = \frac{k(k+1)}{2}$ .

Now

$$\begin{aligned} \sum_{i=1}^{k+1} i &= \left( \sum_{i=1}^k i \right) + (k+1) \\ &= \frac{k(k+1)}{2} + k+1 \quad \{\text{using } P_k\} \\ &= \frac{k(k+1)}{2} + \frac{2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2} \\ &= \frac{(k+1)([k+1] + 1)}{2}. \end{aligned}$$

$\therefore P_{k+1}$  is also true.

Since  $P_1$  is true, and  $P_{k+1}$  is true whenever  $P_k$  is true,  $P_n$  is true for all positive integers  $n$ .      {principle of mathematical induction}

## Question 3

Use the principle of mathematical induction to prove that for all positive integers  $n$ :

$$\sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}.$$

## Answer

Let  $P_n$  be the proposition:  $\sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}$  for all positive integers  $n$ .

**Proof:** (By the principle of mathematical induction)

(1) If  $n = 1$ : LHS =  $1 \times 2 = 2$  and RHS =  $\frac{1(2)(3)}{3} = 2$ .  $\therefore P_1$  is true.

(2) If  $P_k$  is true, then  $\sum_{i=1}^k i(i+1) = \frac{k(k+1)(k+2)}{3}$ .

Now

$$\begin{aligned} \sum_{i=1}^{k+1} i(i+1) &= \left( \sum_{i=1}^k i(i+1) \right) + (k+1)(k+2) \\ &= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2) \quad \{\text{using } P_k\} \\ &= \frac{k(k+1)(k+2) + 3(k+1)(k+2)}{3} \\ &= \frac{(k+1)(k+2)(k+3)}{3} \\ &= \frac{[k+1]([k+1]+1)([k+1]+2)}{3}. \end{aligned}$$

$\therefore P_{k+1}$  is also true.

Since  $P_1$  is true, and  $P_{k+1}$  is true whenever  $P_k$  is true,  $P_n$  is true for all positive integers  $n$ .      {principle of mathematical induction}

## Question 4

Use the principle of mathematical induction to prove that for all positive integers  $n$ :

$$\sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}.$$

## Answer

Let  $P_n$  be the proposition:  $\sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}$  for all positive integers  $n$ .

**Proof:** (By the principle of mathematical induction)

(1) If  $n = 1$ : LHS =  $1^3 = 1$  and RHS =  $\frac{1^2(2)^2}{4} = 1$ .  $\therefore P_1$  is true.

(2) If  $P_k$  is true, then  $\sum_{i=1}^k i^3 = \frac{k^2(k+1)^2}{4}$ . Note: both terms below share a factor of  $(k+1)^2$ , since  $(k+1)^3 = (k+1)(k+1)^2$ , so there is no need to fully expand  $(k+1)^3$ . Now

$$\begin{aligned} \sum_{i=1}^{k+1} i^3 &= \left( \sum_{i=1}^k i^3 \right) + (k+1)^3 \\ &= \frac{k^2(k+1)^2}{4} + (k+1)^3 \quad \{\text{using } P_k\} \\ &= \frac{k^2(k+1)^2 + 4(k+1)^3}{4} \\ &= \frac{(k+1)^2[k^2 + 4(k+1)]}{4} \quad \left\{ (k+1)^3 = (k+1)(k+1)^2 \right\} \\ &= \frac{(k+1)^2(k^2 + 4k + 4)}{4} \\ &= \frac{(k+1)^2(k+2)^2}{4} \\ &= \frac{[k+1]^2([k+1] + 1)^2}{4}. \end{aligned}$$

$\therefore P_{k+1}$  is also true.

Since  $P_1$  is true, and  $P_{k+1}$  is true whenever  $P_k$  is true,  $P_n$  is true for all positive integers  $n$ . {principle of mathematical induction}